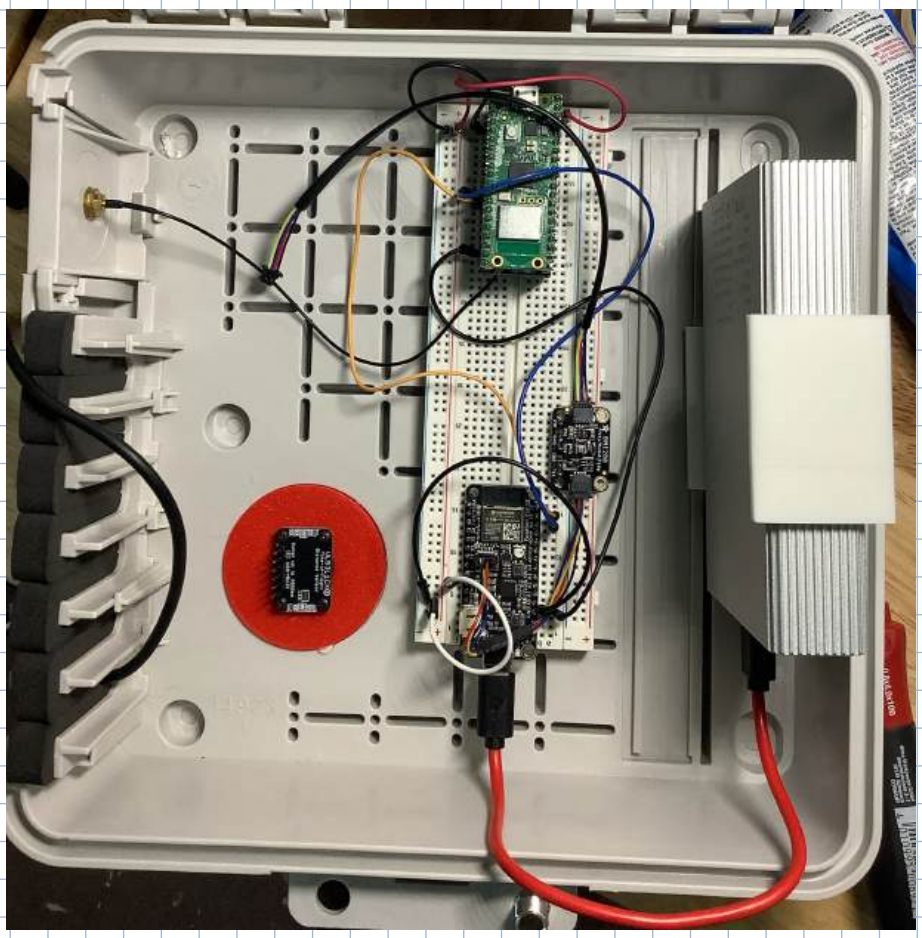


6/21/26 FATHER'S DAY

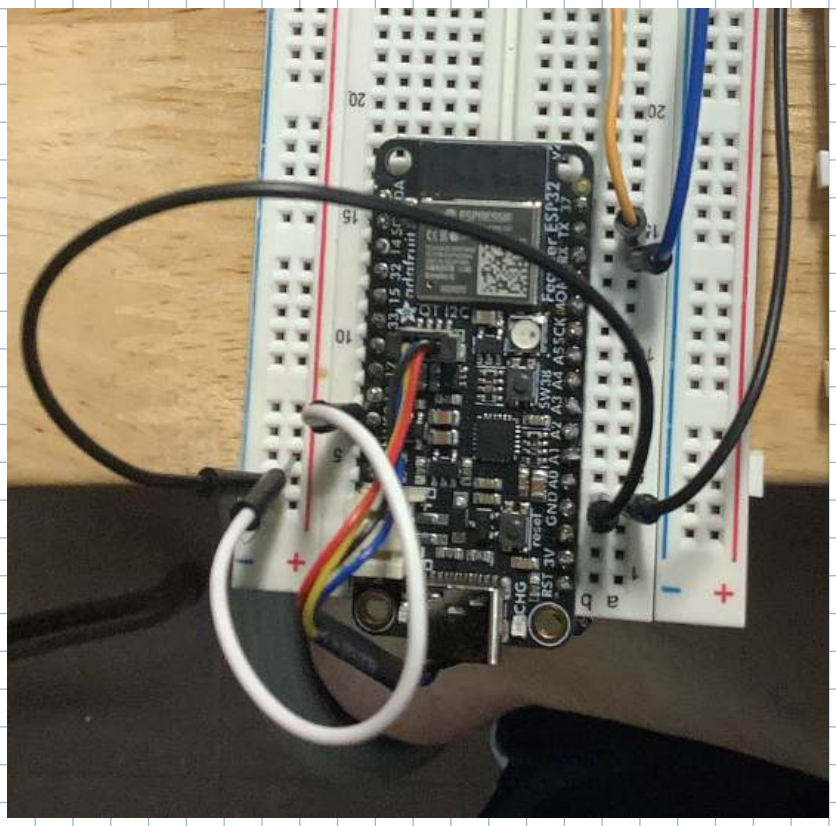
TTD

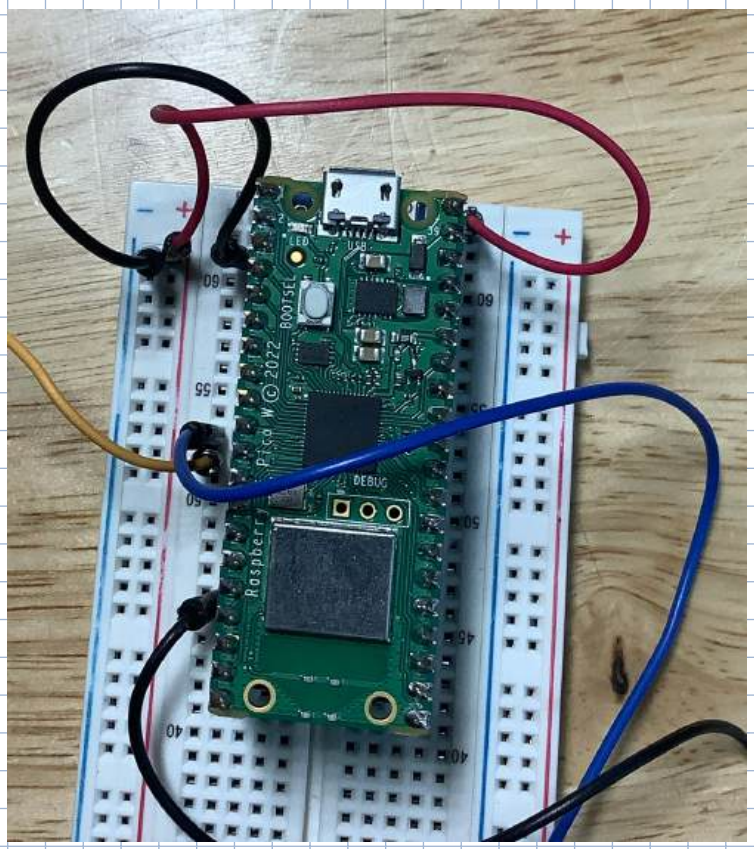
1. RE-WIRE ELECTRONICS ON 2 SHORT BOARDS FOR A BETTER FIT ✓ 6/24
2. TEST SOFTWARE ✓
3. FINISH FIXTURES ✓

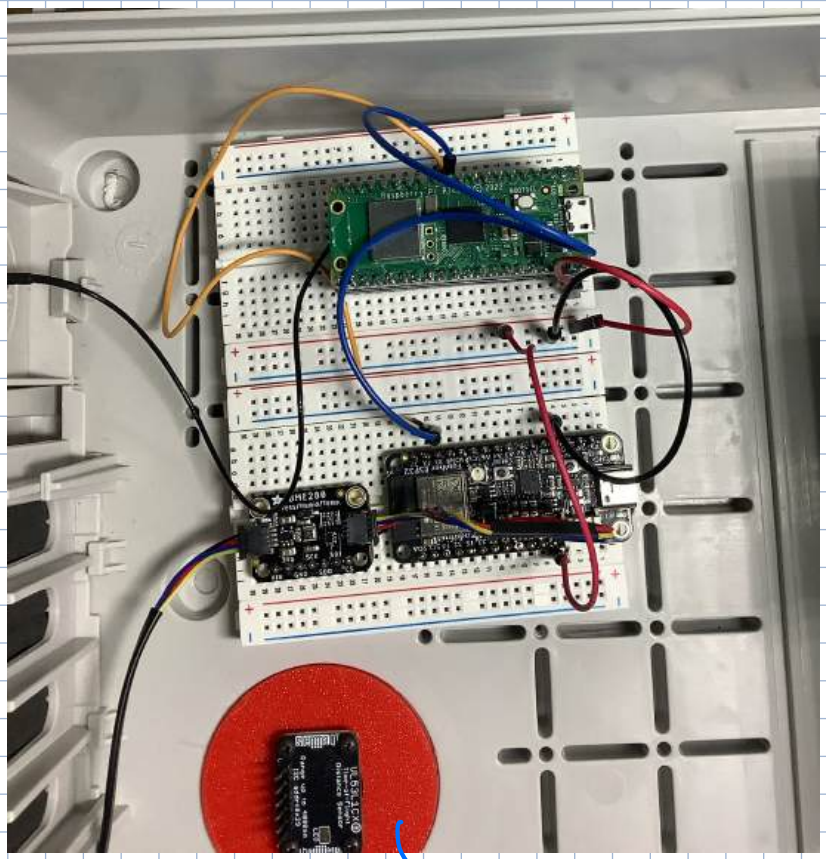
4. TEST IN A TUB OF WATER ✓ 6/24

NOT WORKING!

OPTICAL SENSOR DOESN'T WORK IN PIPE.







Acoustic sensor instead?

REDUCE!

JULY 6 2026

REPLACED LIDAR SENSOR WITH M5 STACK I²C ACOUSTIC SENSOR.

① MATT HELPED ME GET THE CODE WORKING AND 3D PRINTED A NEW SENSOR FITTING

② JULIA WROTE CODE TO PARSE THE MODBUS MESSAGES AND SAVE THE T, P, H & DISTANCE DATA IN AN EXCEL SPREADSHEET USING PYTHON

③ I ASSEMBLED THE NEW VERSION AND TESTED IT WITH A BILLING PIPE IN A TUB OF WATER

(OK! IT WORKS! ULTRASONIC MEASURED DISTANCE WITHIN AN INCH OF TAPE MEASURE)

NOW GET PHOTOS & DEPLOY THIS SOMEWHERE!

FIX UP CODE - GET STANDARD DEVIATION TOO!

IF DRAW IS 100 mA ON AVERAGE, HOW MUCH ENERGY IS NEEDED PER DAY?

$5V \cdot 100\,mA \cdot 3600 \cdot 24$

$5 \cdot 100 \times 10^{-3} \cdot 3600 \cdot 24\,Hr$

$5 \cdot 100 \cdot 3.6$

.5 W · 24 Hr

6W SOLAR CELL $\frac{1}{2}$ DAY $\frac{1}{2}$ ENERGY ABSORBED · 24 Hr

1.5 W 12 Hr

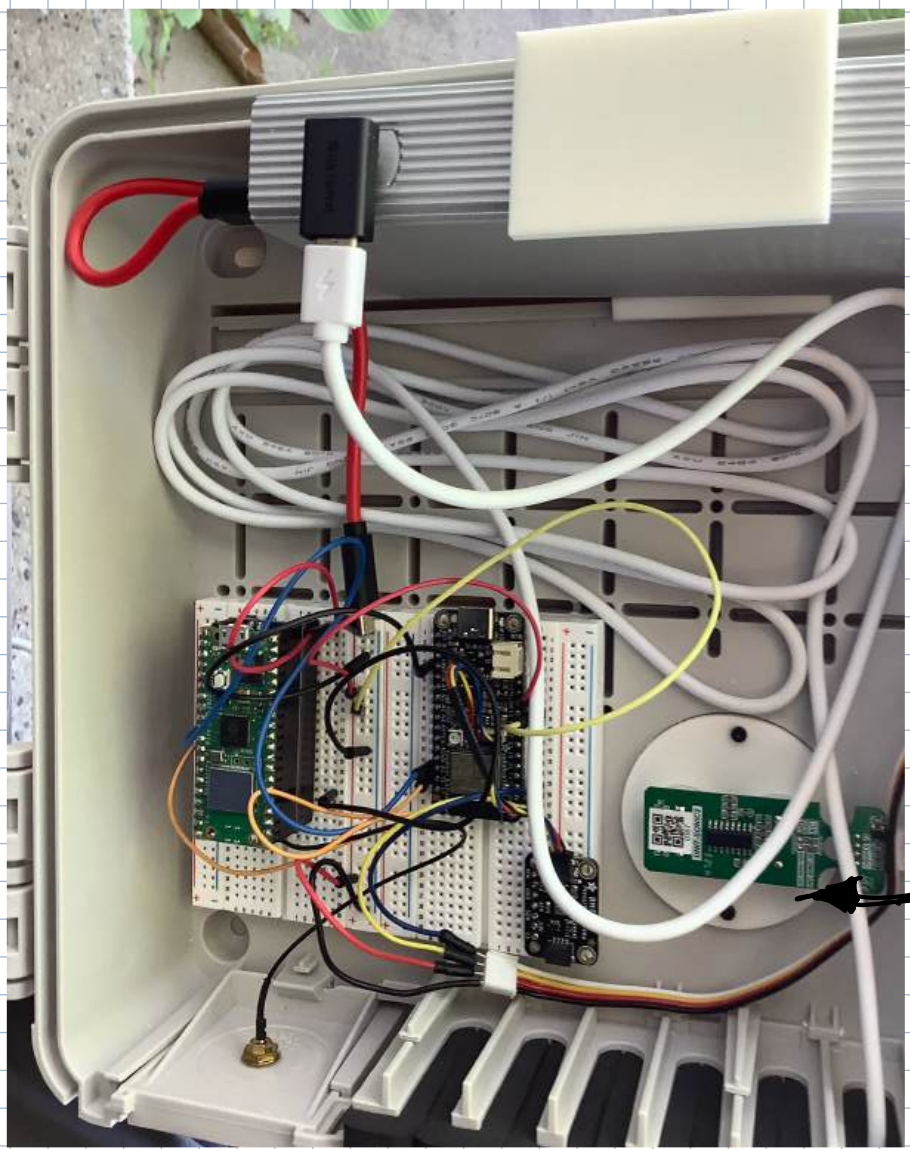
I THINK THIS SHOULD BE OK THROUGHOUT THE YEAR

JULY 8 2026









NEW ACOUSTIC SEASON

JULY 8 2026

REPLACED LIDAR WITH ACOUSTIC SENSOR

M5 STACK I²C

MATT 3-D PRINTED FIXTURE FOR SENSOR

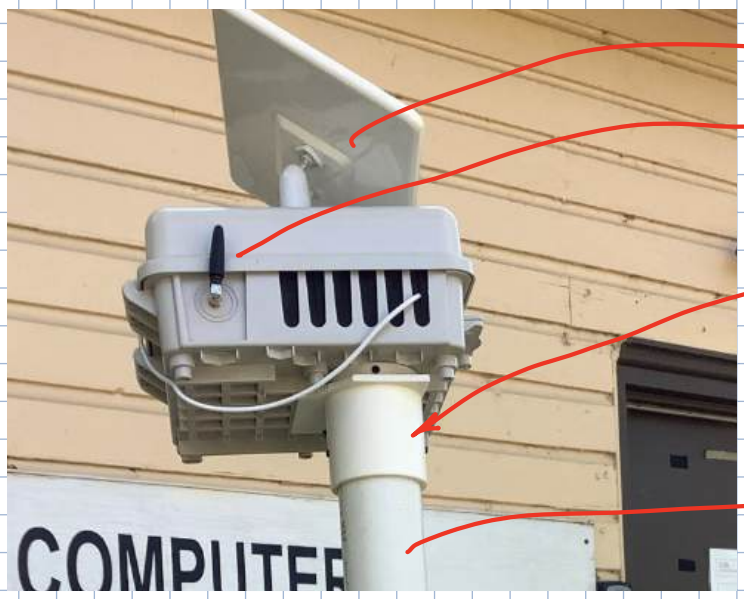
OK! SENSOR SEEMS TO WORK! HOWEVER THE AVERAGE VALUE CHANGES RANDOMLY. CODE HAS BEEN MODIFIED TO SEND OUT ALL 10 DATA POINTS, NOT JUST AVERAGE

↳ !! SOMETIMES SENSOR SENDS 4500 mm !!
(IF YOU IGNORE THESE READINGS, THINGS ARE VERY CONSISTENT!)
(MAXIMUM VALUE)

NEED TO MODIFY CODE TO IGNORE BAD READINGS

BRING UNIT TO FOSH OPEN HOUSE ON SATURDAY AND SEE IF WE CAN FIND A PLACE FOR IT

ADAFRUIT HAS A FEATHER WING LATCHING RELAY TO TURN OFF PERIPHERALS TO SAVE MORE ENERGY
↳ NEXT STEP



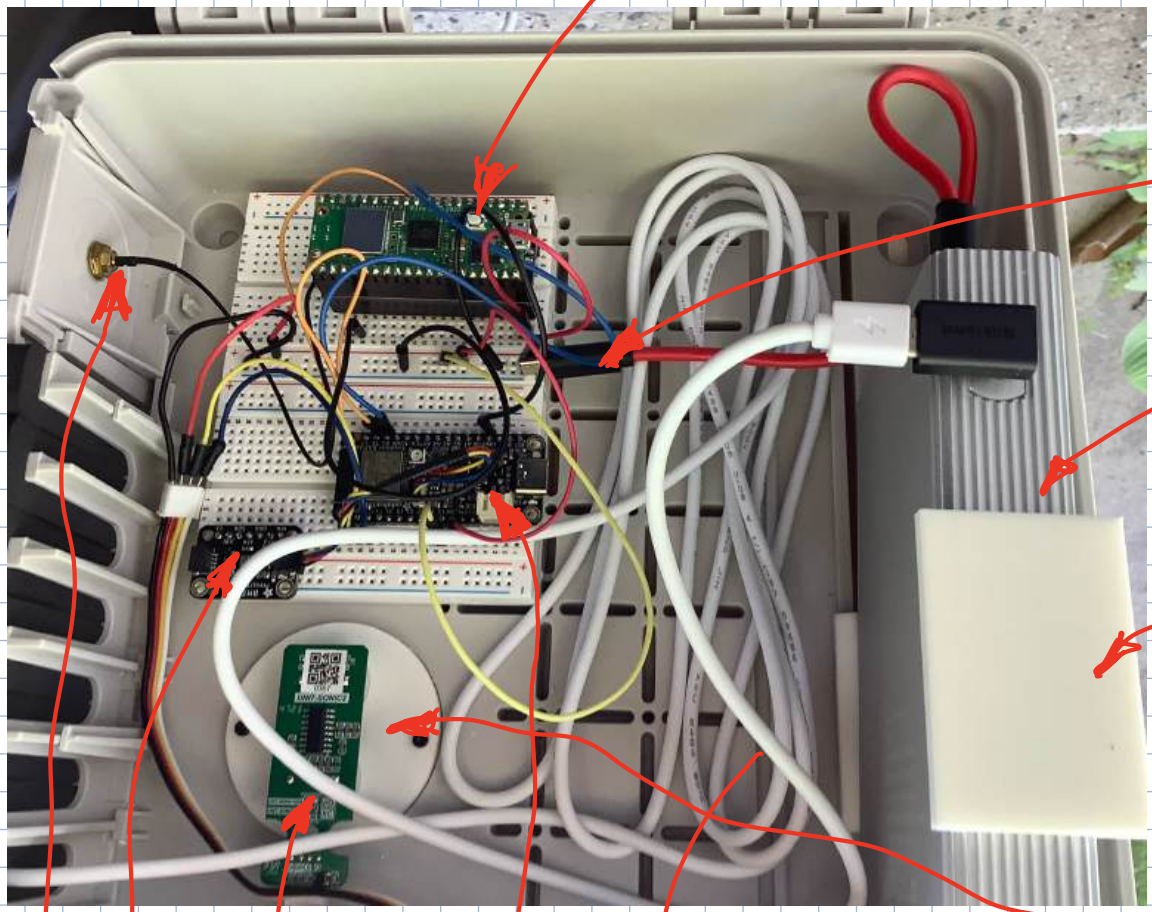
6W SOLAR PANEL

MESHTASTIC ANTENNA

3D PRINTED STILLING PIPE SUPPORT

2" PVC PIPE

PICO W + MESHTASTIC TRANSMITTER



FEATHER M0 W32 (DISCONNECTED)

BATTERY

3D PRINTED BATTERY HOLDER

3D PRINTED SENSOR FIXTURE

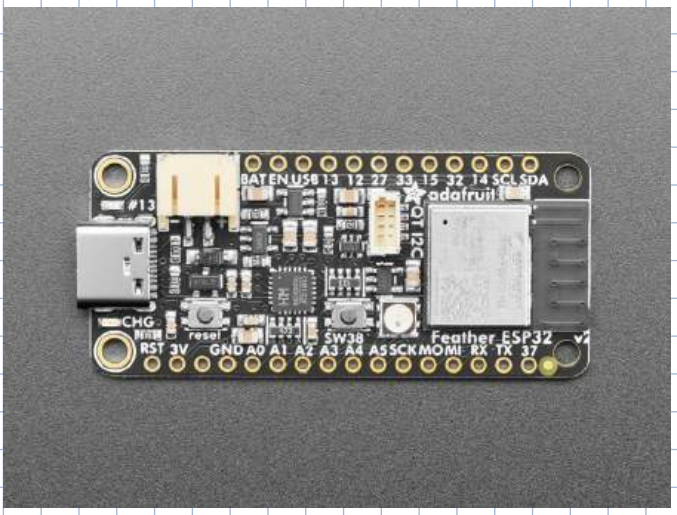
M5 STACK I2C ACOUSTIC SENSOR

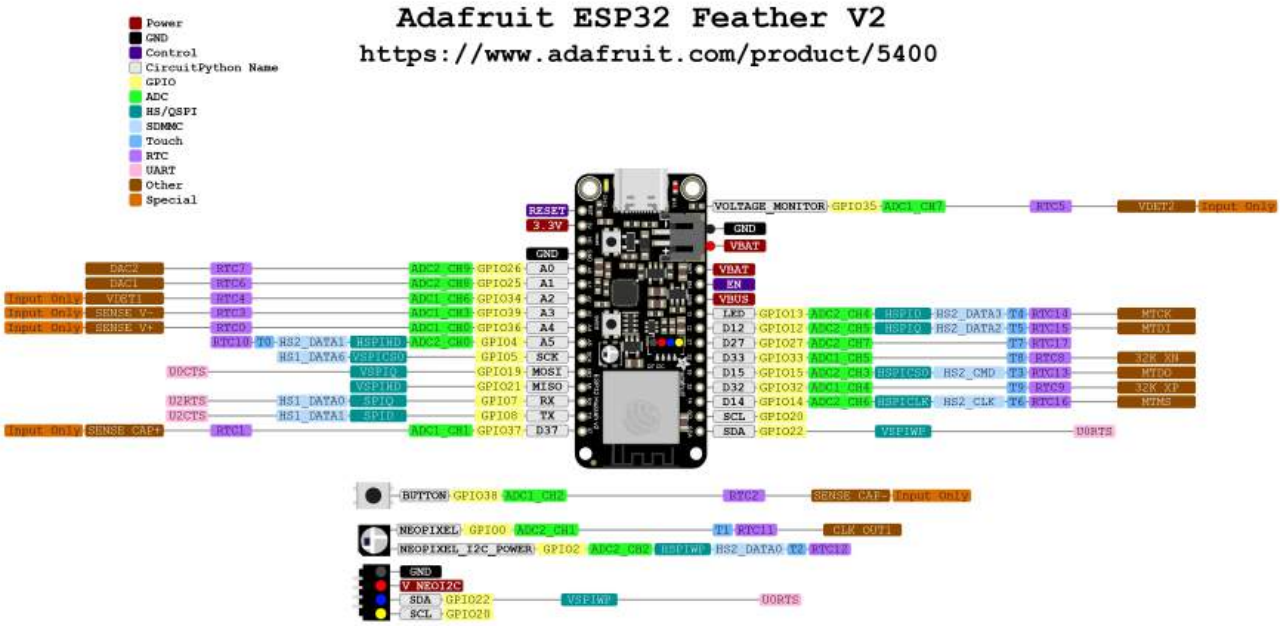
CABLE TO 6W SOLAR PANEL

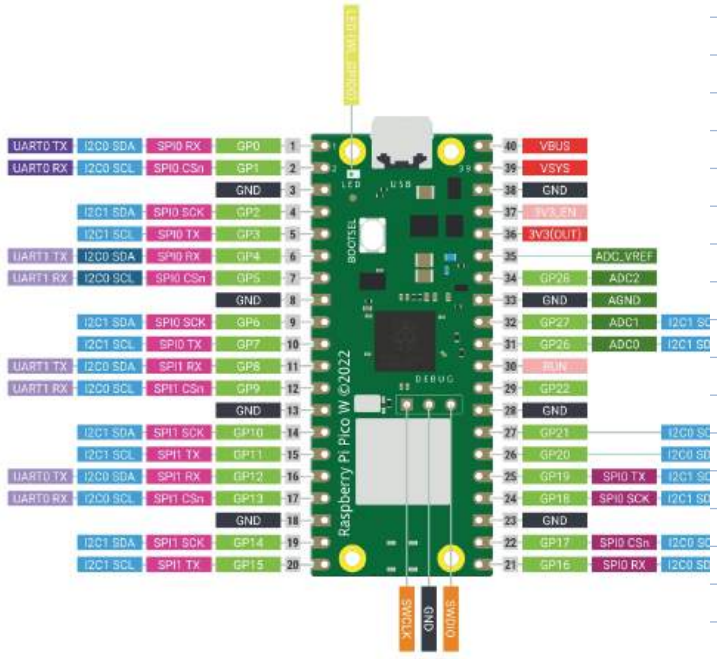
BME 280 T, P, H SENSOR

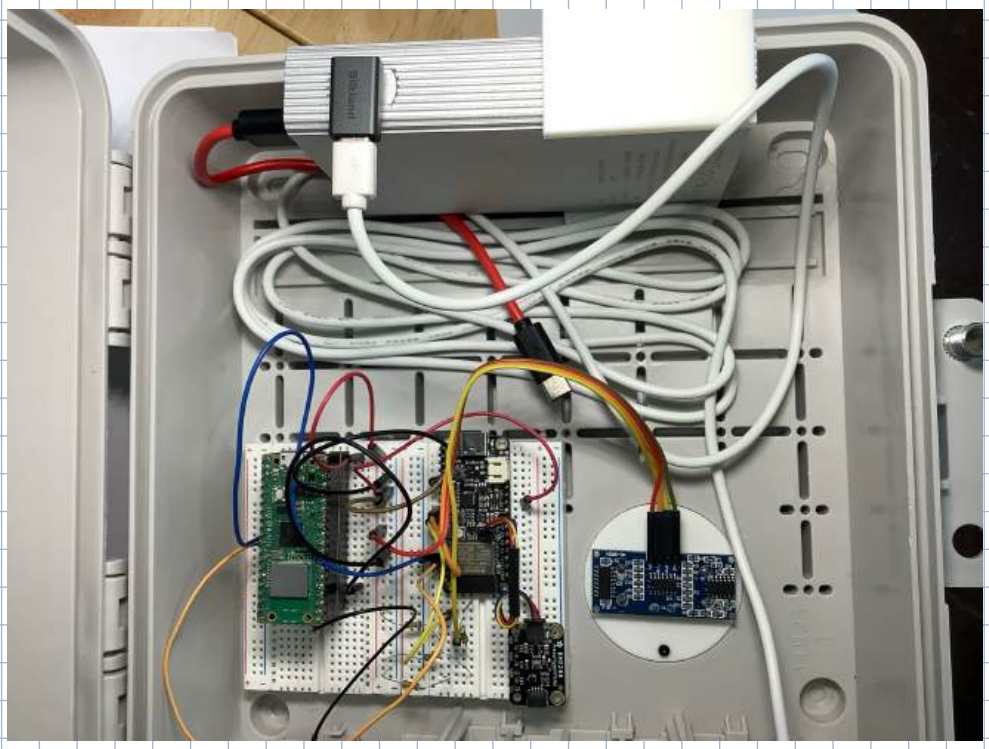
ESP32 V2 FEATHER

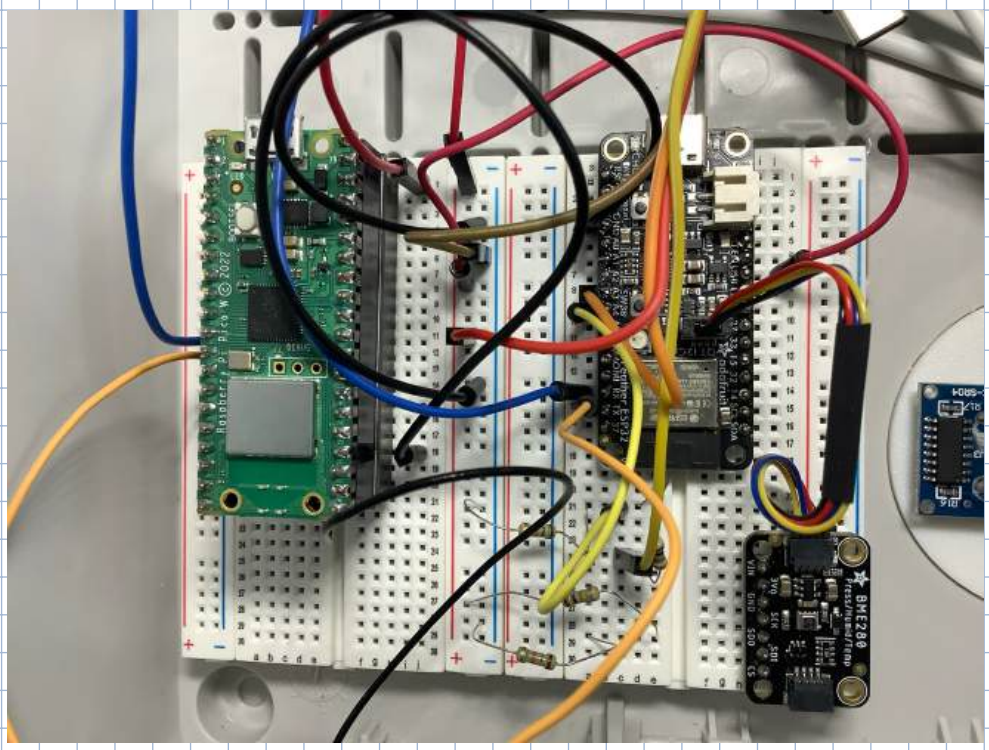
MESHTASTIC ANTENNA


BAT EN USB 13 12 27 33 15 32 14 SCL SDA
adafruit
QT I2C
CHG
reset
SW38
Feather ESP32 v2
RST 3V GND A0 A1 A2 A3 A4 A5 SCK MOMI RX TX 37


Adafruit ESP32 Feather V2
https://www.adafruit.com/product/5400
Power
GND
Control
CircuitPython Name
GPIO
ADC
HS/QSPI
SDMMC
Touch
RTC
UART
Other
Special
RESET
3.3V
GND
VBAT
EN
VBUS
A0 GPIO26
A1 GPIO25
A2 GPIO34
A3 GPIO39
A4 GPIO36
A5 GPIO04
SCK GPIO05
MOSI GPIO19
MISO GPIO21
RX GPIO07
TX GPIO08
D37 GPIO37
LED GPIO13
D12 GPIO12
D27 GPIO27
D33 GPIO33
D15 GPIO15
D32 GPIO32
D14 GPIO14
SCL GPIO20
SDA GPIO22
VOLTAGE_MONITOR GPIO35
BUTTON GPIO38
NEOPIXEL GPIO00
NEOPIXEL_I2C_POWER GPIO02
V_NEOI2C
SDA GPIO22
SCL GPIO20


Raspberry Pi Pico W ©2022
BOOTSEL
DEBUG
SWCLK
GND
SWDIO




BME280

UPDATE AUGUST 2 2026

I FOUND OUT WHY THE TIDE SENSOR WASN'T WORKING. I FORGOT TO DEFINE PINMODES FOR THE ULTRASONIC SENSOR!!

```
pinMode(TRIG_PIN, OUTPUT)
pinMode(ECHO_PIN, INPUT)
```

I ALSO SWAPPED OUT ON HC-SR04 SENSOR FOR ANOTHER (NOT THE PROBLEM) AND I SWITCHED TO PORTS FROM 36, 39 TO 14, 15 (NOT THE PROBLEM EITHER) I SCREWED UP THE AVERAGE VALUE TOO! - CORRECTED

TidesensorV5 ARDUINO SCRIPT HAS BEEN UPDATED AND THINGS LOOK OK!!

IF THE PIPE IS NOT INSTALLED THE SENSOR GIVES SPURIOUS READINGS (127 MM), BUT WITH THE PIPE AND A BUCKET OF WATER, THINGS LOOK GOOD!

MECHATASTIC PORTION SEEMS TO BE WORKING WELL TOO.

OUTSIDE IN PAIL OF WATER

MEASURED DISTANCE = 80 INCHES (2032 MM) (WITH TAPE MEASURE)

(1) READING = 1986 MM

BUCKET OF WATER 72 INCHES (72 3/8"?) 1829 MM

(2) READING = 1809 MM

OK!!

1808 M
1807

(3) ADD WATER FROM PAIL TO BUCKET

TAPE = 70" 1778 MM
SENSOR = 1752 MM
1751 MM
1752 MM

④ ~~SHAFTS~~ CAPTED SENSOR UP

TAPE 72.5" (1840 mm)
SENSOR 1822 MM
1824 MM

③ TAPE 78 3/4" (2000 mm)
79" (2007 mm)
SENSORS 1956
1951

⑥ TAPE 76 1/4" (1937 mm)
1887 MM
1886
1885

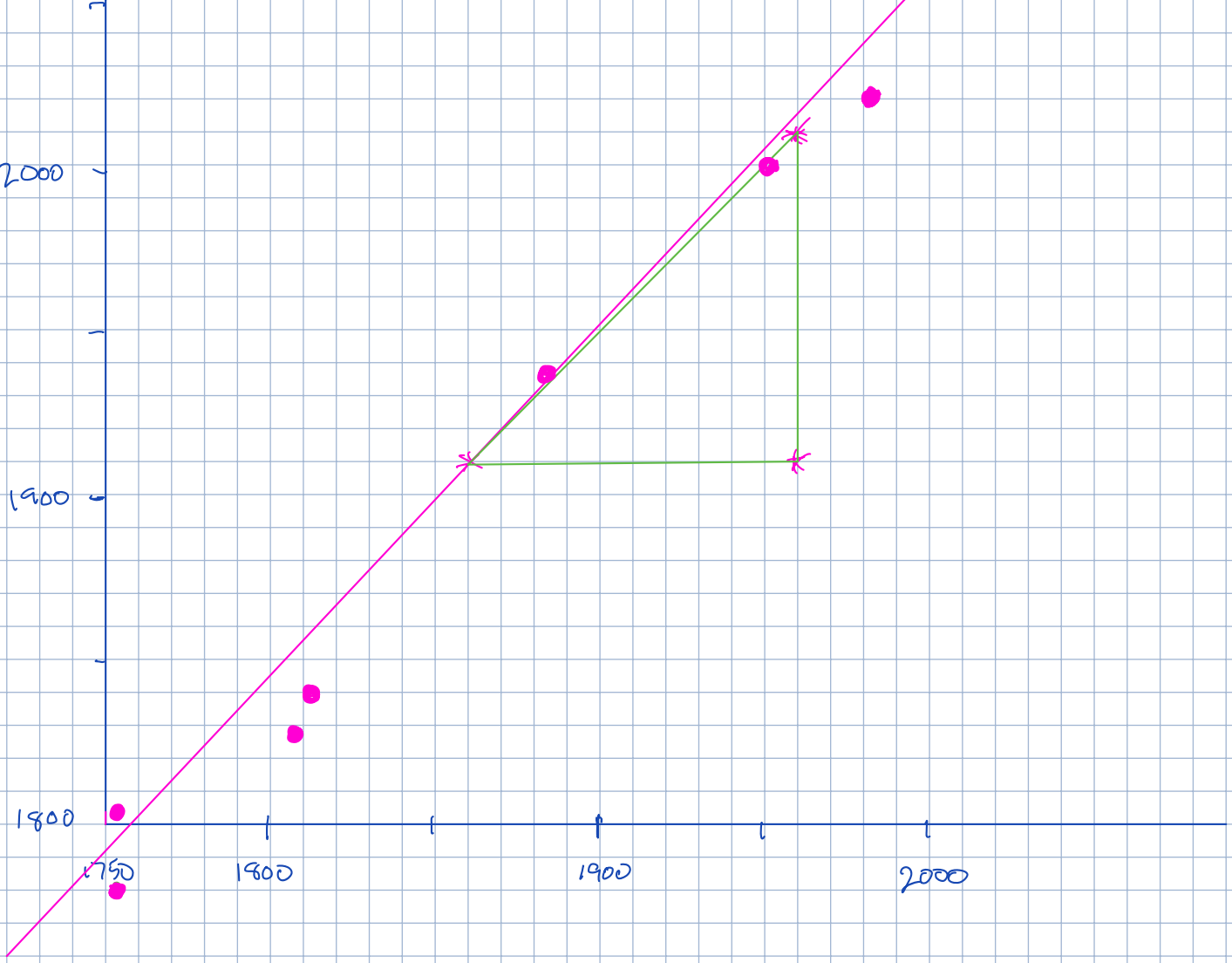
⑦ 81" (2057 mm)
1962
1962

⑧ 71" (1803 mm)
1753



OK! I THINK EVERYTHING IS WORKING. I LEFT THE SYSTEM ON TO SEE IF I CAN SEE A SIGNAL AT HOME.

OK! GOT ONE DISTANCE MEASUREMENT FROM HOME (FRONT PORCH)



THINGS TO FINISH

8/5 ✓
- ✓ BOTTOM PAINT INSIDE OF PIPE
- ✓ TEST IN NORMAL MODE NOT TEST MODE
- ○ FIGURE OUT HOW TO COLLECT DATA WITH TIMESTAMP
- ○ ANALYSIS SUN & MOON POSITIONS
- ~○ PERF BOARD & RELAY CIRCUIT ORDERED BOARDS ARRIVED
- ○ 3D PRINT LABEL => TIDE SENSOR INFO AGE COL
  - WAITING FOR EMAIL FROM NEIL @

8/6 ✓
- ✓ REASSEMBLE & TEST IN BUCKET @ CDL
- ✗ GET UNIVERSITY PAINT TO MARK POSITION ON POLE
- ✓ INSTALL AT FOBH
- ○

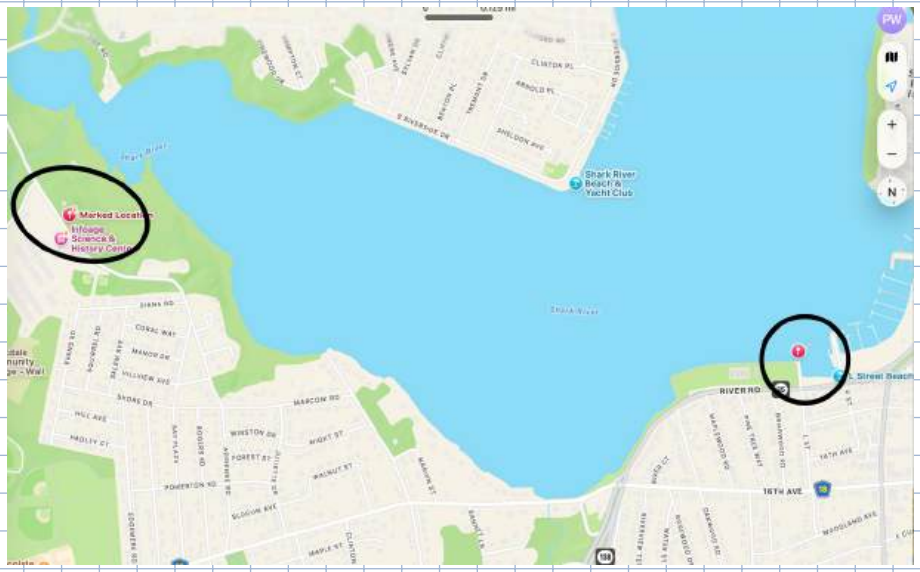
OK! DEPLOYED



OK! PICKED UP SIGNAL ON PICNIC BENCH @ FOBH AND GOT MESSAGES AT HOME TOO!

~10 AM

10:52



~ 1½ hr AFTER LOW TIDE

~10:25 AM 1890 MM

~ 1:55 PM 945 MM

OK!!

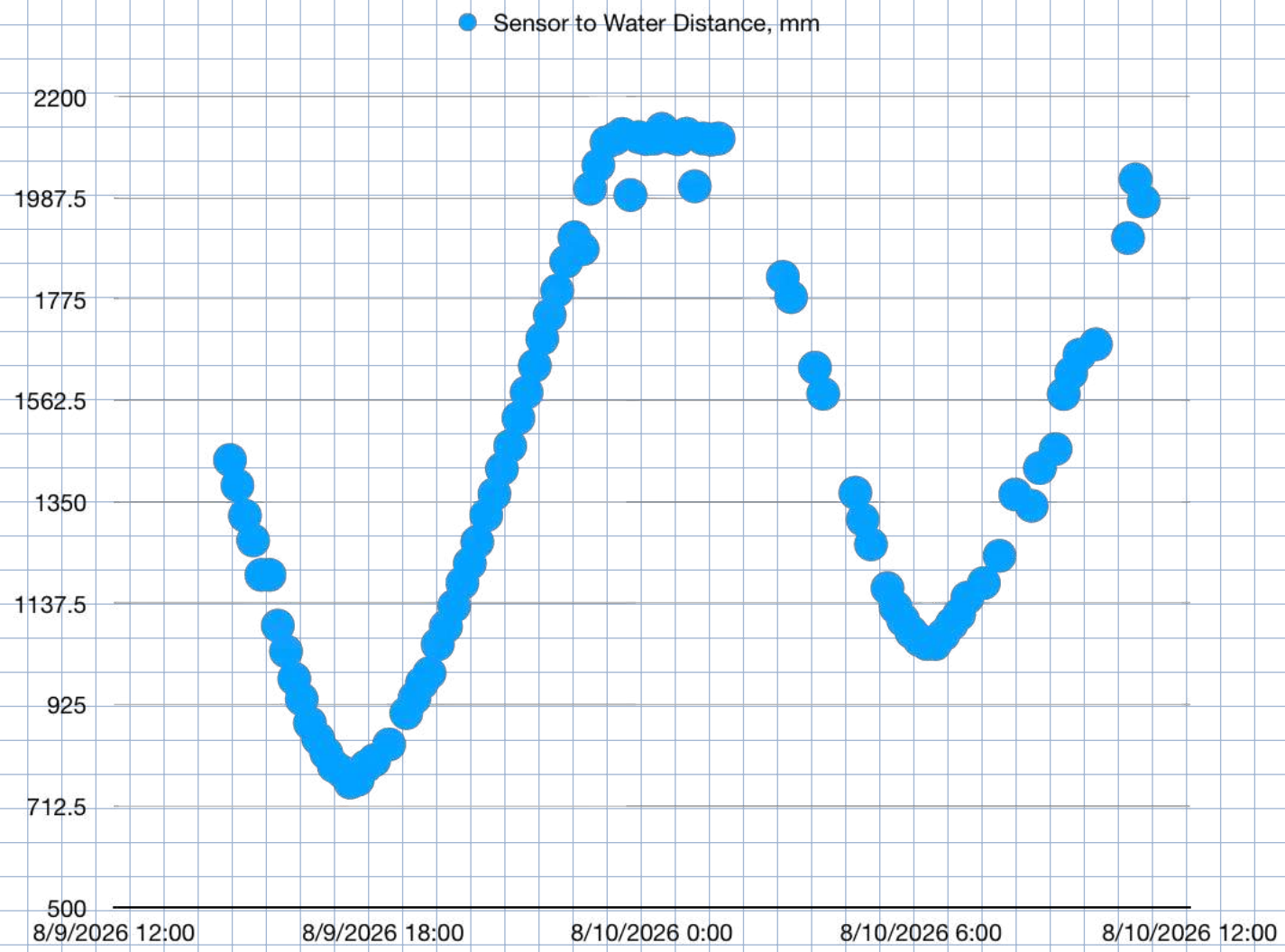
~3 ft CHANGE!

LOST DATA, BUT DEFINITELY SAW SMOOTH DECREASING TREND AS TIDE ROSE!

UPDATE 8/6/26

TIDE SENSOR IS SHOWING > ½ METER TIDE DIFFERENCE THROUGHOUT THE DAY (THUR & FRIDAY)

NEXT STEP IS TO COLLECT DATA



8/17/26

OH NO! SENSOR IS GONE!

STRAPS ARE STILL THERE, BUT BOX + PIPE + SPACERS ARE LOST!